Báo cáo tích hợp — Đặc tả → Kripke → Sinh mã → CWE/CVE

Đặc tả: smart_ev_static_fuzz_experiment

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1. Đặc tả yêu cầu (Specification)

GD9.3 experiment: adds Ch2 (buffer_copy for token handling) and Ch4 (divide for charging rate) semantic adapters. V2 state machine preserved. SMT checks preserved + corrected overflow probe added. All mappings are semantically honest — no fake functions.

Mã yêu cầu	Loại	Ưu tiên	Nội dung
REQ-AUTH-01	safety	HIGH	See Phase 1 audit and Phase 6 property suite for REQ-AUTH-01.
REQ-AUTH-02	safety	HIGH	See Phase 1 audit and Phase 6 property suite for REQ-AUTH-02.
REQ-SOC-01	safety	HIGH	See Phase 1 audit and Phase 6 property suite for REQ-SOC-01.
REQ-SOC-02	safety	HIGH	See Phase 1 audit and Phase 6 property suite for REQ-SOC-02.
REQ-TEMP-01	safety	HIGH	See Phase 1 audit and Phase 6 property suite for REQ-TEMP-01.
REQ-PORT-01	safety	HIGH	See Phase 1 audit and Phase 6 property suite for REQ-PORT-01.
REQ-PORT-02	safety	HIGH	See Phase 1 audit and Phase 6 property suite for REQ-PORT-02.
REQ-PORT-03	safety	HIGH	See Phase 1 audit and Phase 6 property suite for REQ-PORT-03.
REQ-CHG-01	safety	HIGH	See Phase 1 audit and Phase 6 property suite for REQ-CHG-01.
REQ-CHG-02	safety	HIGH	See Phase 1 audit and Phase 6 property suite for REQ-CHG-02.
REQ-EMG-01	safety	HIGH	See Phase 1 audit and Phase 6 property suite for REQ-EMG-01.

2. Kiểm chứng hình thức (Kripke/CTL)

Trạng thái deadlock: không có

Công thức CTL	Kết quả	Trạng thái thỏa
AG(!req_auth_01_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation
AG(!req_auth_02_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation

Công thức CTL	Kết quả	Trạng thái thỏa
AG(!req_soc_01_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation
AG(!req_soc_02_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_temp_01_violation

Công thức CTL	Kết quả	Trạng thái thỏa
AG(!req_temp_01_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation
AG(!req_port_01_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation

Công thức CTL	Kết quả	Trạng thái thỏa
AG(!req_port_02_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation
AG(!req_port_03_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation

Công thức CTL	Kết quả	Trạng thái thỏa
AG(!req_chg_01_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation
AG(!req_chg_02_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation

Công thức CTL	Kết quả	Trạng thái thỏa
AG(!req_emg_01_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation
EF(charging)	ĐÚNG	AUTHENTICATED, CHARGING, CONNECTED, DISCONNECTED, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT
EF(start_blocked)	ĐÚNG	AUTH_FAILED, CONNECTED, DISCONNECTED, START_BLOCKED
EF(stopped)	ĐÚNG	AUTHENTICATED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL
EF(recovered)	ĐÚNG	AUTHENTICATED, CHARGING, CONNECTED, DISCONNECTED, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT
AF(safe)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, RECOVERED_SAFE, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL

2.1. Sơ đồ chuyển trạng thái

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 1/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 1/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=blue!25] (nd_DISCONNECTED) at (0.36,2.15) {DISCONNECTED\\{\tiny
auth\_clear,disconnected,power\_zero,safe}};
\node[fill=gray!15] (nd_CONNECTED) at (-0.22,0.37) {CONNECTED\\{\tiny
connected,not\_authenticated,port\_occupied,power\_zero,safe}};
\node[fill=gray!15] (nd_AUTH_FAILED) at (-0.38,-0.77) {AUTH\_FAILED\\{\tiny
auth\_failed,connected,power\_zero,safe,start\_not\_allowed}};
\node[fill=gray!15] (nd_AUTHENTICATED) at (-2.20,0.66) {AUTHENTICATED\\{\tiny
authenticated,connected,port\_ready,power\_zero,safe}};
\node[fill=gray!15] (nd_START_BLOCKED) at (1.61,1.07) {START\_BLOCKED\\{\tiny
connected,power\_zero,safe,start\_blocked}};
\node[fill=gray!15] (nd_CHARGING) at (-0.70,-0.19) {CHARGING\\{\tiny
authenticated,charging,connected,power\_positive,safe,safety\_gate\_valid}};
\node[fill=gray!15] (nd_RECOVERED_SAFE) at (1.20,-1.00) {RECOVERED\_SAFE\\{\tiny
authenticated,connected,power\_zero,recovered,safe}};
\node[fill=gray!15] (nd_OVERHEAT) at (-0.92,-1.78) {OVERHEAT\\{\tiny
authenticated,connected,overheat,stop\_required,unsafe\_until\_stopped}};
\node[fill=gray!15] (nd_PORT_FAULT) at (-0.21,1.86) {PORT\_FAULT\\{\tiny
authenticated,connected,fault,port\_fault,stop\_required,unsafe\_until\_stopped}};
\node[fill=gray!15] (nd_DISCONNECTING) at (0.55,-1.58) {DISCONNECTING\\{\tiny
auth\_clear\_required,disconnecting,stop\_required}};
\node[fill=gray!15] (nd_SOC_FULL) at (0.32,0.09) {SOC\_FULL\\{\tiny
authenticated,connected,soc\_full,stop\_required,unsafe\_until\_stopped}};
\node[fill=gray!15] (nd_EMERGENCY) at (-1.38,-1.27) {EMERGENCY\\{\tiny
emergency,stop\_required,unsafe\_until\_stopped}};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__STOPPED_PORT_FAULT) at (1.23,1.00) {$\to$
STOPPED\_PORT\_FAULT};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__STOPPED_OVERHEAT) at (-0.66,-0.67) {$\to$ STOPPED\_OVERHEAT};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__STOPPED_EMERGENCY) at (0.86,-1.84) {$\to$
STOPPED\_EMERGENCY};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__STOPPED_DISCONNECT) at (1.79,-0.09) {$\to$
STOPPED\_DISCONNECT};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__STOPPED_SOC_FULL) at (-1.25,1.99) {$\to$ STOPPED\_SOC\_FULL};
\draw (nd_DISCONNECTED) -- (nd_DISCONNECTED);
\draw (nd_DISCONNECTED) -- (nd_CONNECTED);
\draw (nd_CONNECTED) -- (nd_AUTH_FAILED);
\draw (nd_CONNECTED) -- (nd_CONNECTED);
\draw (nd_CONNECTED) -- (nd_AUTHENTICATED);
\draw (nd_AUTH_FAILED) -- (nd_START_BLOCKED);
\draw (nd_AUTH_FAILED) -- (nd_AUTH_FAILED);
\draw (nd_AUTHENTICATED) -- (nd_AUTHENTICATED);
\draw (nd_AUTHENTICATED) -- (nd_CHARGING);
\draw (nd_START_BLOCKED) -- (nd_START_BLOCKED);
\draw (nd_CHARGING) -- (nd_OVERHEAT);
\draw (nd_CHARGING) -- (nd_PORT_FAULT);
\draw (nd_CHARGING) -- (nd_DISCONNECTING);
\draw (nd_CHARGING) -- (nd_SOC_FULL);
\draw (nd_CHARGING) -- (nd_EMERGENCY);
\draw (nd_CHARGING) -- (nd_CHARGING);
\draw (nd_RECOVERED_SAFE) -- (nd_RECOVERED_SAFE);
\draw (nd_RECOVERED_SAFE) -- (nd_CHARGING);
\draw (nd_OVERHEAT) -- (nd_OVERHEAT);
\draw (nd_OVERHEAT) -- (nd___ghost__STOPPED_OVERHEAT);
\draw (nd_PORT_FAULT) -- (nd___ghost__STOPPED_PORT_FAULT);
\draw (nd_PORT_FAULT) -- (nd_PORT_FAULT);
\draw (nd_DISCONNECTING) -- (nd___ghost__STOPPED_DISCONNECT);
```

```
\draw (nd_DISCONNECTING) -- (nd_DISCONNECTING);
\draw (nd_SOC_FULL) -- (nd___ghost__STOPPED_SOC_FULL);
\draw (nd_SOC_FULL) -- (nd_SOC_FULL);
\draw (nd_EMERGENCY) -- (nd_EMERGENCY);
\draw (nd_EMERGENCY) -- (nd___ghost__STOPPED_EMERGENCY);
\draw (nd___ghost__STOPPED_PORT_FAULT) -- (nd_RECOVERED_SAFE);
\draw (nd___ghost__STOPPED_OVERHEAT) -- (nd_RECOVERED_SAFE);
\draw (nd___ghost__STOPPED_EMERGENCY) -- (nd_RECOVERED_SAFE);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 2/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 2/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_STOPPED_EMERGENCY) at (0.13,1.98) {STOPPED\_EMERGENCY\\{\tiny
emergency,power\_zero,safe,stopped}};
\node[fill=gray!15] (nd_STOPPED_PORT_FAULT) at (1.07,0.71) {STOPPED\_PORT\_FAULT\\{\tiny
authenticated,connected,fault,power\_zero,safe,stopped}};
\node[fill=gray!15] (nd_STOPPED_OVERHEAT) at (-1.49,-1.16) {STOPPED\_OVERHEAT\\{\tiny
authenticated,connected,overheat,power\_zero,safe,stopped}};
\node[fill=gray!15] (nd_STOPPED_DISCONNECT) at (-2.13,0.52) {STOPPED\_DISCONNECT\\{\tiny
auth\_clear,disconnected,power\_zero,safe,stopped}};
\node[fill=gray!15] (nd_STOPPED_SOC_FULL) at (1.76,1.36) {STOPPED\_SOC\_FULL\\{\tiny
authenticated,connected,power\_zero,safe,soc\_full,stopped}};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__RECOVERED_SAFE) at (-1.21,1.74) {$\to$ RECOVERED\_SAFE};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__EMERGENCY) at (2.13,-0.02) {$\to$ EMERGENCY};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__PORT_FAULT) at (0.46,-0.89) {$\to$ PORT\_FAULT};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__OVERHEAT) at (-2.01,-0.85) {$\to$ OVERHEAT};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__DISCONNECTING) at (-0.91,-2.07) {$\to$ DISCONNECTING};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__SOC_FULL) at (2.20,-1.32) {$\to$ SOC\_FULL};
\draw (nd_STOPPED_EMERGENCY) -- (nd_STOPPED_EMERGENCY);
\draw (nd_STOPPED_EMERGENCY) -- (nd___ghost__RECOVERED_SAFE);
\draw (nd_STOPPED_PORT_FAULT) -- (nd___ghost__RECOVERED_SAFE);
\draw (nd_STOPPED_PORT_FAULT) -- (nd_STOPPED_PORT_FAULT);
\draw (nd_STOPPED_OVERHEAT) -- (nd_STOPPED_OVERHEAT);
\draw (nd_STOPPED_OVERHEAT) -- (nd___ghost__RECOVERED_SAFE);
\draw (nd_STOPPED_DISCONNECT) -- (nd_STOPPED_DISCONNECT);
\draw (nd_STOPPED_SOC_FULL) -- (nd_STOPPED_SOC_FULL);
\draw (nd___ghost__EMERGENCY) -- (nd_STOPPED_EMERGENCY);
\draw (nd___ghost__PORT_FAULT) -- (nd_STOPPED_PORT_FAULT);
\draw (nd___ghost__OVERHEAT) -- (nd_STOPPED_OVERHEAT);
\draw (nd___ghost__DISCONNECTING) -- (nd_STOPPED_DISCONNECT);
\draw (nd___ghost__SOC_FULL) -- (nd_STOPPED_SOC_FULL);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 3/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 3/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_chg_02_violation) at (0.00,0.00) {unreachable\_req\_chg\_02\_violation\\{\tiny
req\_chg\_02,req\_chg\_02\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_chg_02_violation) -- (nd_unreachable_req_chg_02_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 4/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 4/13
```

```
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_port_03_violation) at (0.00,0.00) {unreachable\_req\_port\_03\_violation\\{\tiny
req\_port\_03,req\_port\_03\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_port_03_violation) -- (nd_unreachable_req_port_03_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 5/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 5/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_port_01_violation) at (0.00,0.00) {unreachable\_req\_port\_01\_violation\\{\tiny
req\_port\_01,req\_port\_01\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_port_01_violation) -- (nd_unreachable_req_port_01_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 6/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 6/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_temp_01_violation) at (0.00,0.00) {unreachable\_req\_temp\_01\_violation\\{\tiny
req\_temp\_01,req\_temp\_01\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_temp_01_violation) -- (nd_unreachable_req_temp_01_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 7/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 7/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_auth_01_violation) at (0.00,0.00) {unreachable\_req\_auth\_01\_violation\\{\tiny
req\_auth\_01,req\_auth\_01\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_auth_01_violation) -- (nd_unreachable_req_auth_01_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 8/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 8/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_port_02_violation) at (0.00,0.00) {unreachable\_req\_port\_02\_violation\\{\tiny
req\_port\_02,req\_port\_02\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_port_02_violation) -- (nd_unreachable_req_port_02_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 9/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 9/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_soc_02_violation) at (0.00,0.00) {unreachable\_req\_soc\_02\_violation\\{\tiny req\_soc\_02,req\_soc\_02\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_soc_02_violation) -- (nd_unreachable_req_soc_02_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 10/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 10/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_soc_01_violation) at (0.00,0.00) {unreachable\_req\_soc\_01\_violation\\{\tiny req\_soc\_01,req\_soc\_01\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_soc_01_violation) -- (nd_unreachable_req_soc_01_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 11/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 11/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_auth_02_violation) at (0.00,0.00) {unreachable\_req\_auth\_02\_violation\\{\tiny req\_auth\_02,req\_auth\_02\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_auth_02_violation) -- (nd_unreachable_req_auth_02_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 12/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 12/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_chg_01_violation) at (0.00,0.00) {unreachable\_req\_chg\_01\_violation\\{\tiny req\_chg\_01,req\_chg\_01\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_chg_01_violation) -- (nd_unreachable_req_chg_01_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_static_fuzz_experiment — phần 13/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_static_fuzz_experiment — phần 13/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
```

\node[fill=gray!15] (nd_unreachable_req_emg_01_violation) at (0.00,0.00) {unreachable_req_emg_01_violation\\{\tiny req_emg_01,req_emg_01_violation,smart_ev_v2,unreachable_violation_witness}};
\draw (nd_unreachable_req_emg_01_violation) -- (nd_unreachable_req_emg_01_violation);
\end{tikzpicture}

3. Mã C sinh ra & Static Checker (lỗi bộ nhớ)

Mã C sinh từ đặc tả

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

/* sinh từ đặc tả 'smart_ev_static_fuzz_experiment', hàm 'handle_auth_token', kind='buffer_copy' */
void handle_auth_token(const char *token_input) {
char buffer[32];
strncpy(buffer, token_input, sizeof(buffer) - 1);
buffer[sizeof(buffer) - 1] = '\0';
printf("%s\n", buffer);
}
```

Không phát hiện lỗi bộ nhớ nào trên mã C sinh ra.

4. Trích xuất AST/điều kiện & SMT (Z3)

Hàm	Biến	Điều kiện	Vòng lặp	Độ phức tạp McCabe	Độ sâu lồng nhau
handle_auth_token	['buffer']	[]	[]	1	1
compute_charging_rate	[]	[]	[]	1	1

Kiểm tra	Kết quả SMT	Model (nếu SAT)
req_soc_01_valid_domain	sat	{'soc': 4}
req_soc_01_violation_unsat	unsat	-
req_soc_01_overflow_check (int32 overflow?)	sat	{'soc_current': -1, 'soc_delta': -2147483648}
req_temp_01_overheat_threshold	sat	{'temperature': 60}
req_temp_01_violation_unsat	unsat	-
req_temp_01_safe_range	sat	{'temperature': 59}
req_soc_01_overflow_v2_bounded	sat	{'soc_current': 0, 'soc_delta': 0}
req_soc_01_overflow_v2_bounded (int32 overflow?)	sat	{'soc_current': -1, 'soc_delta': -2147483648}

5. Dynamic Fuzzing (hàm Python sinh ra)

Hàm Python 'compute_charging_rate'

```
def compute_charging_rate(energy_kwh, time_hours):
return energy_kwh // time_hours
```

Hàm	Kết quả fuzzing
compute_charging_rate	black-box: 11/300 lan crash

6. Phân loại CWE/CVE tổng hợp

Nguồn	Vị trí	Loại phát hiện	CWE	Ví dụ CVE minh họa
ch3	req_soc_01_overflow_check	integer-overflow	CWE-190 — Integer Overflow or Wraparound	CVE-2016-5195 (2016)
ch3	req_soc_01_overflow_v2_bounded	integer-overflow	CWE-190 — Integer Overflow or Wraparound	CVE-2016-5195 (2016)
ch4	compute_charging_rate	uncontrolled-crash	CWE-248 — Uncaught Exception	-